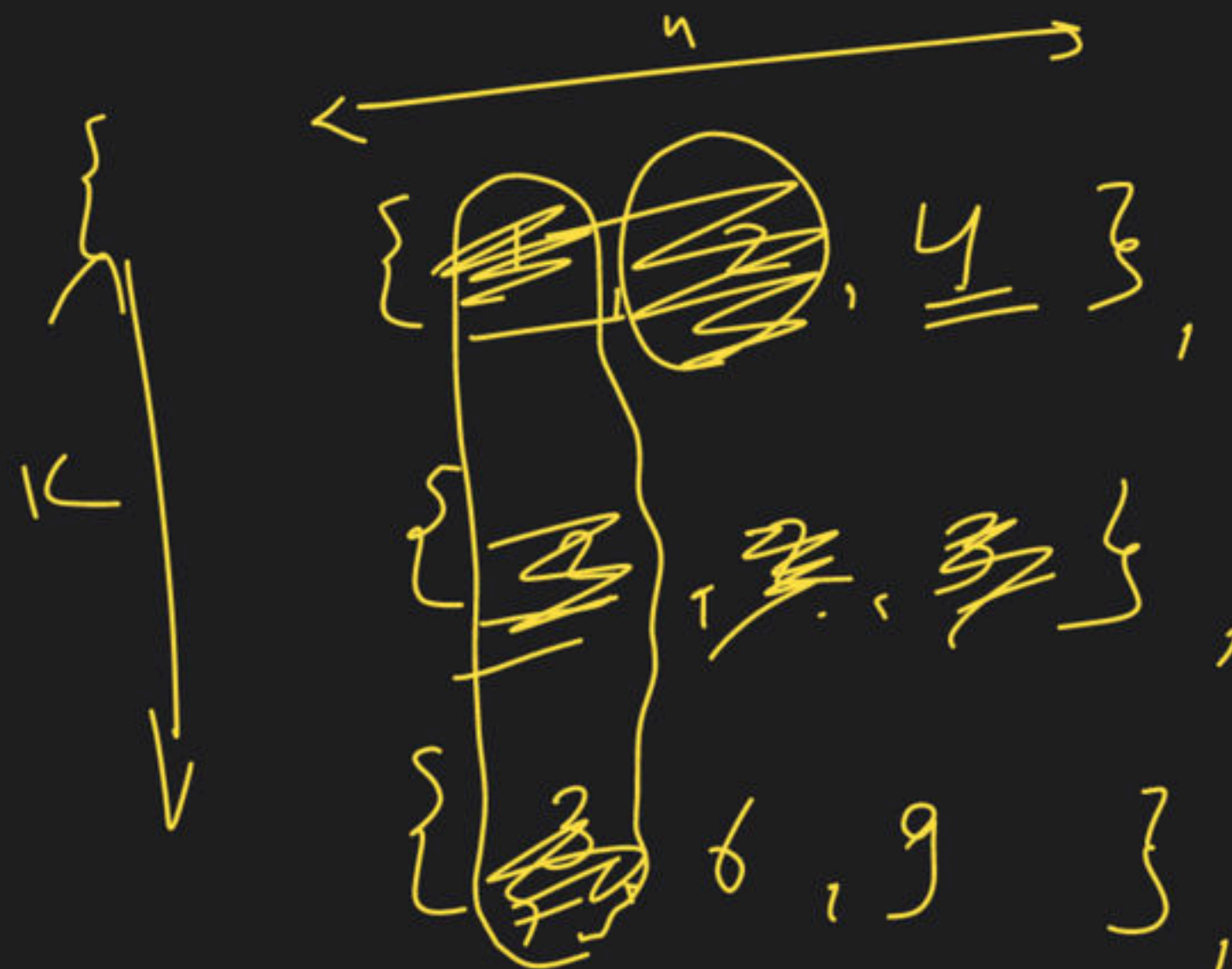


Heaps Class - 4

Special class

→ if ->



#, single array → 1011
↳ sort

$$T. (n \cdot K (\log(n \cdot K)))$$

set was

1, 2, 3

4, 5, 6

2, 2, 3

4, 6

4, 2, 3

4, 2, 3

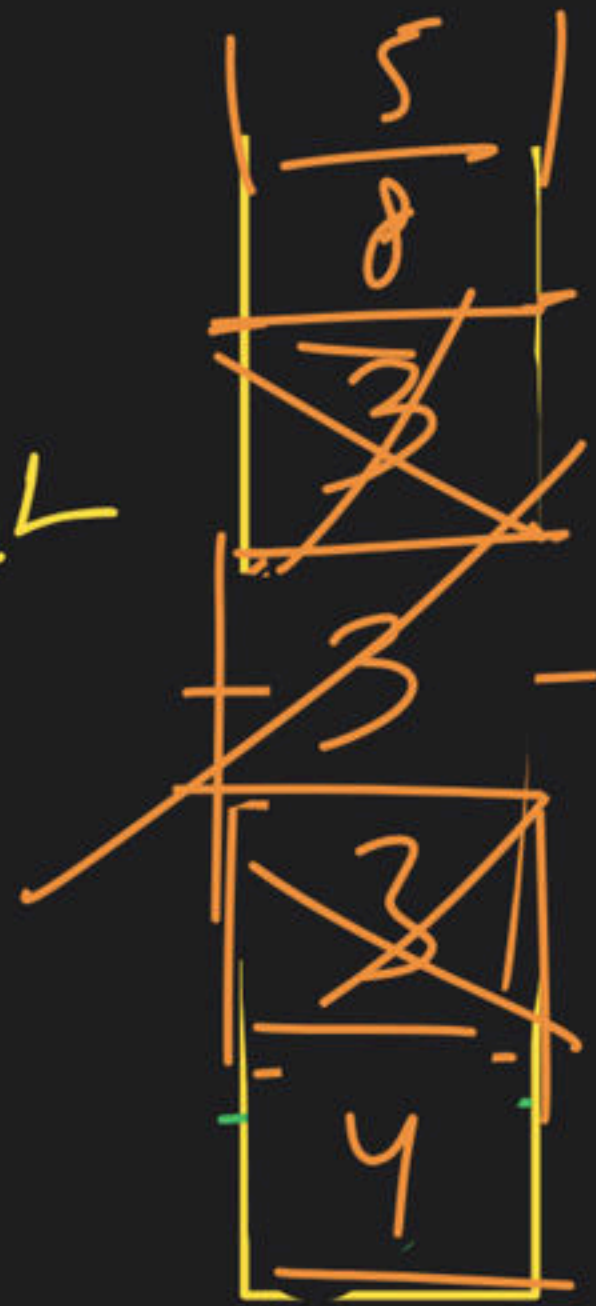
4, 3, 3

6, 9

1	2	2	2	3	3	4	6	9
---	---	---	---	---	---	---	---	---

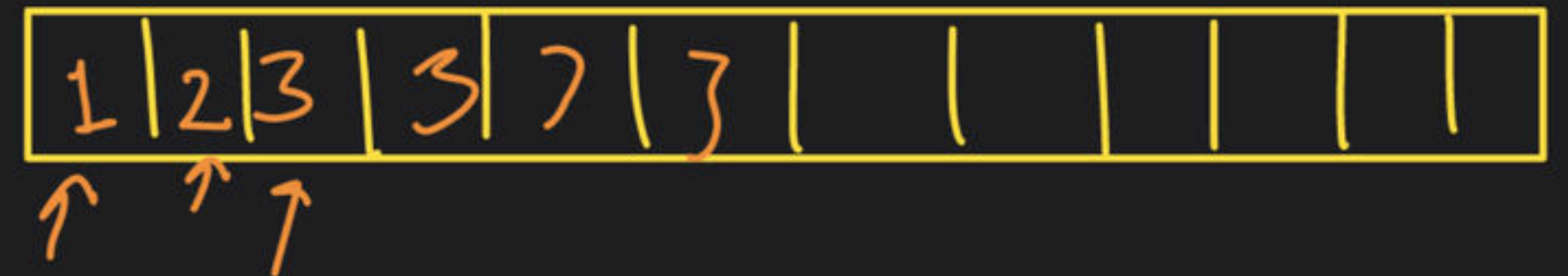
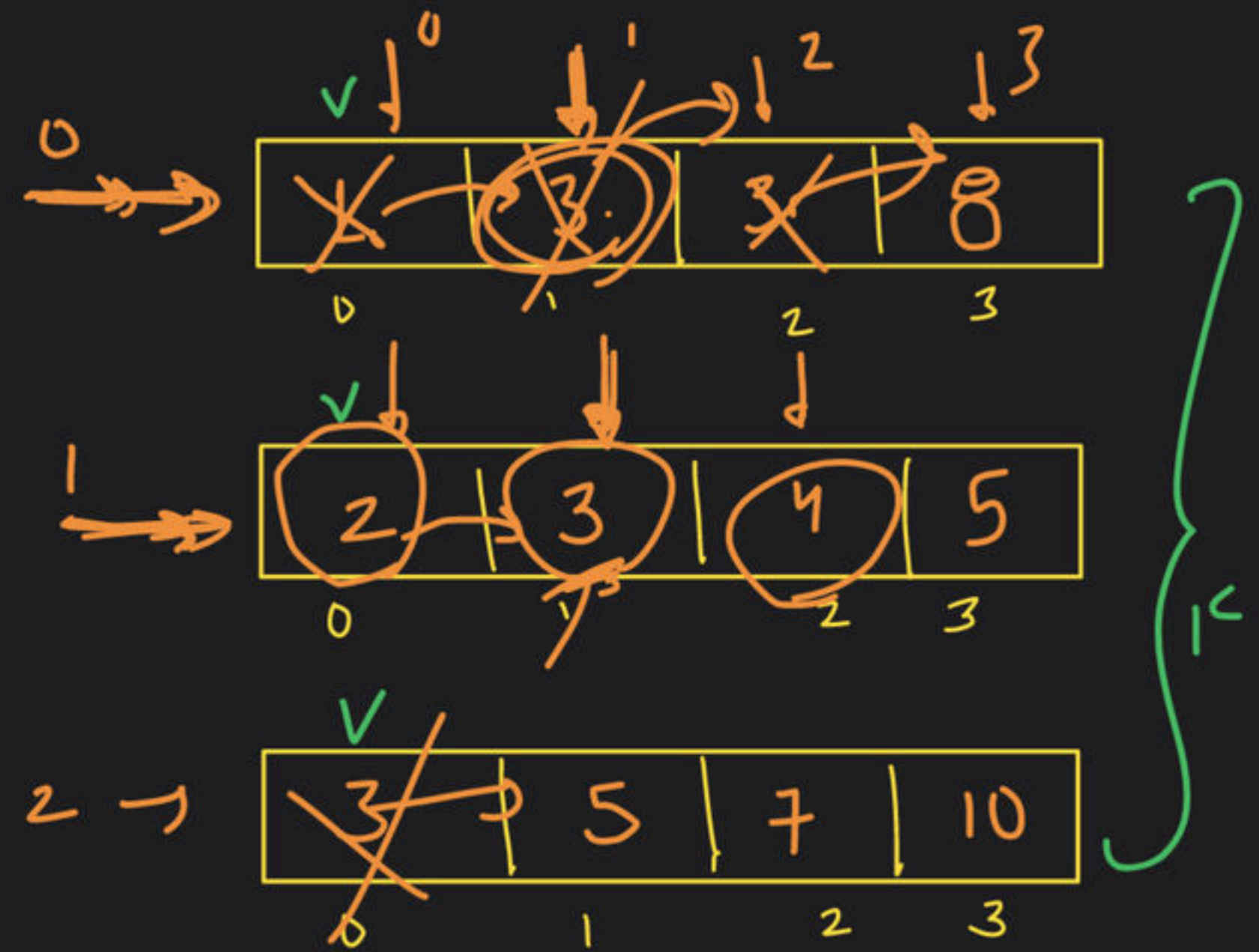
① create heap

② Fix k places
prev



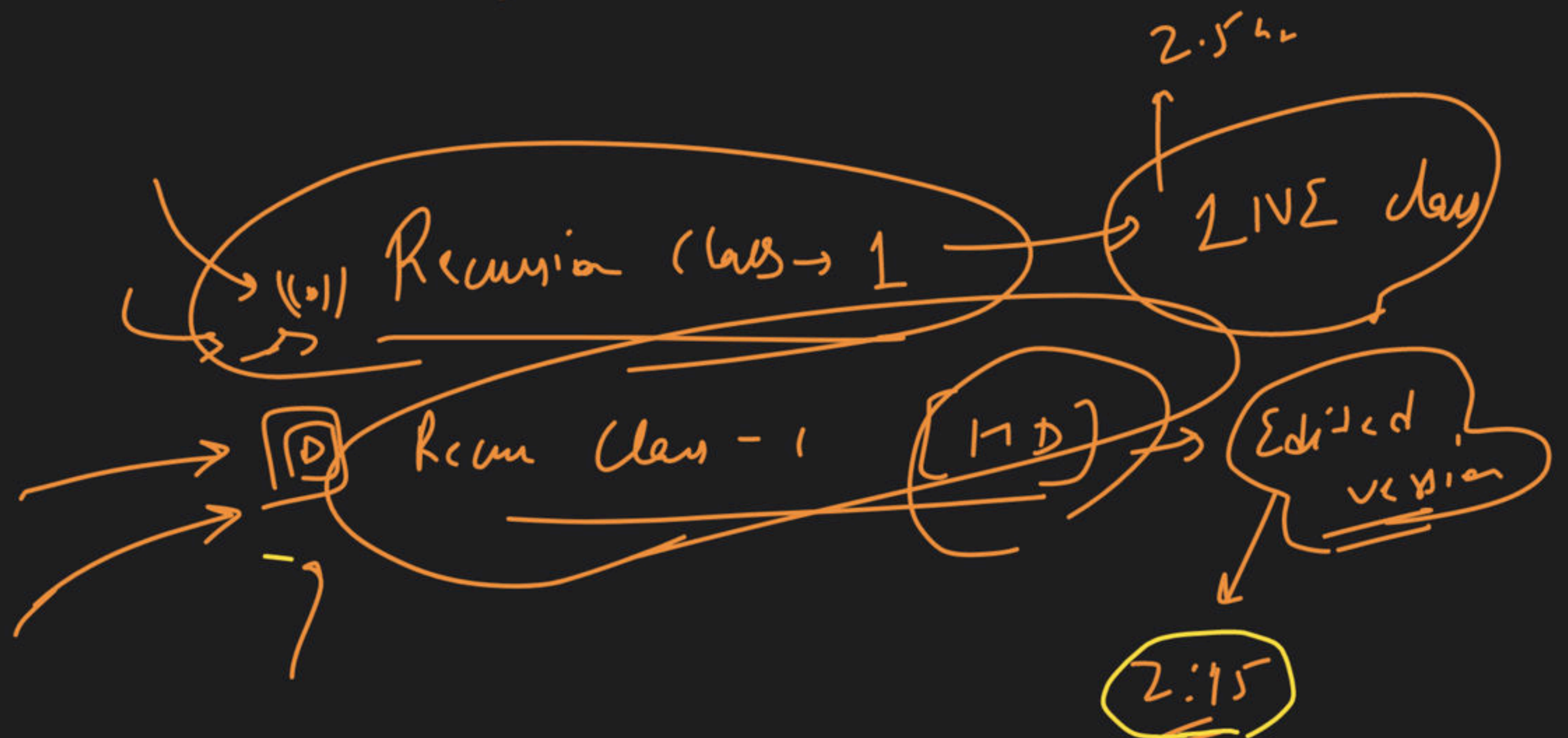
③ main Logic

- ↳ Front data
- ↳ ans insert
- ↳ after while element check



→ Merge K Sorted L-L

BST



20



① create heap

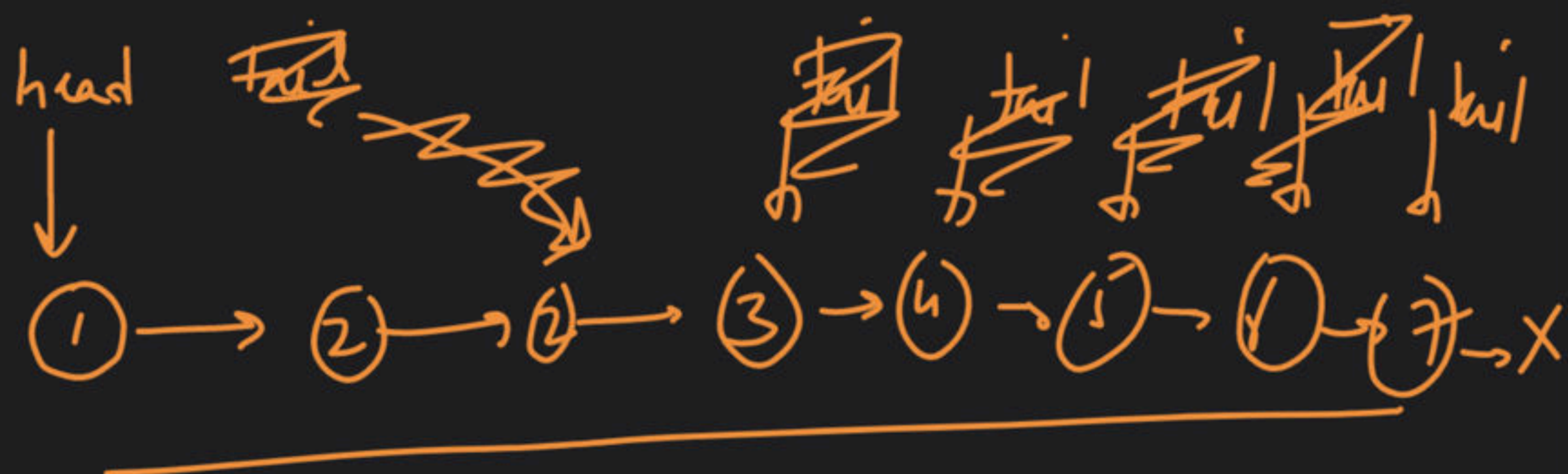
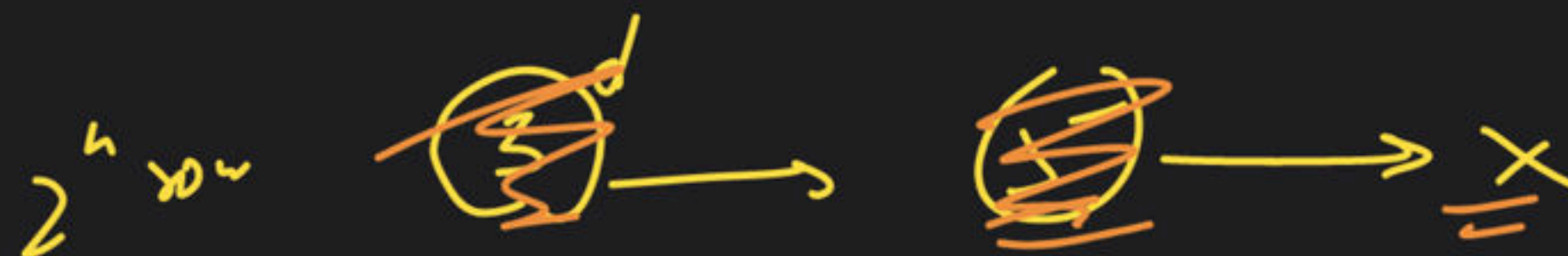
② K - dist
first



③ main logic

↳ front
↳ ans store

↳ agar ka element insert



→ Smallest Range in K List

$\bar{5} \rightarrow 0 \rightarrow 5$
 $\bar{1} \rightarrow 4 \rightarrow 9$
 $\bar{5} \rightarrow 5 \rightarrow 10$
 $\bar{9} \rightarrow 9 \rightarrow 18$
 $\checkmark 10 \rightarrow 18$

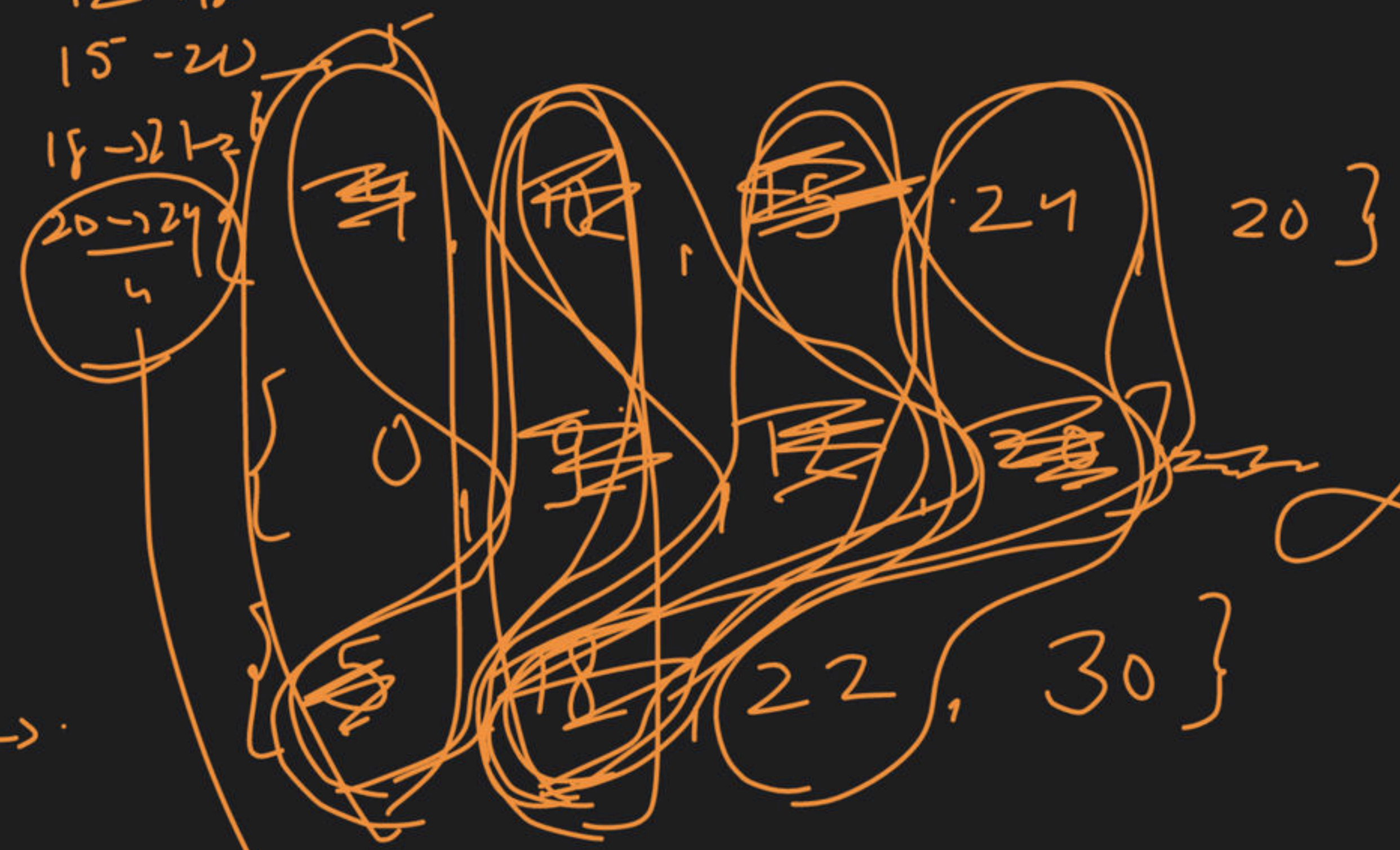
$\text{min} \rightarrow 0$
 $\text{max} \rightarrow 5$

~~1~~
 $\bar{4}$
 $\bar{5}$

$4 \rightarrow$

$12 \rightarrow 18 \rightarrow 6$
 $15 \rightarrow 20$
 $18 \rightarrow 21 \rightarrow 3$
 $20 \rightarrow 24$
 4

$x \rightarrow y$
 $[x, y]$

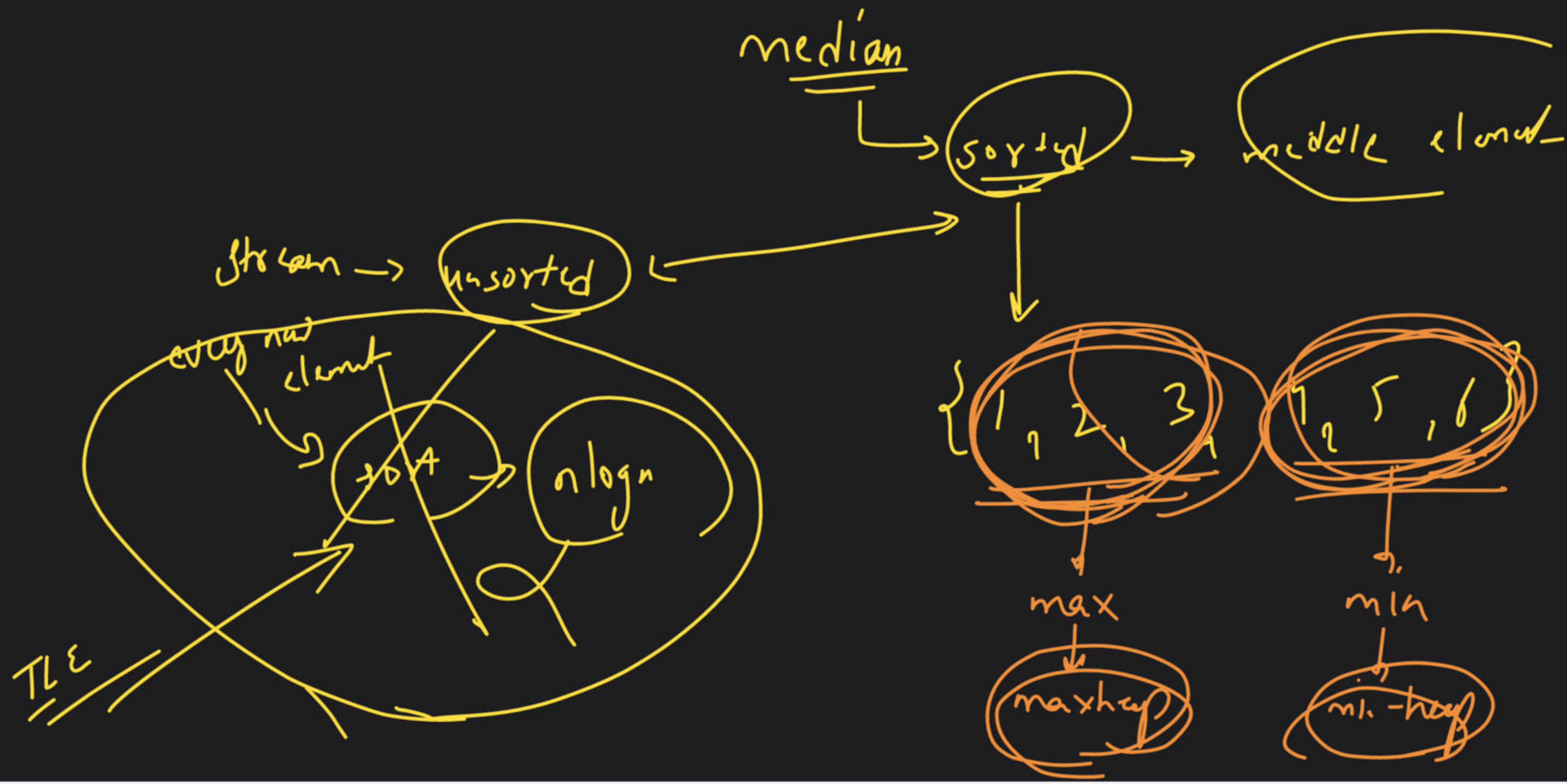


yuk gyk
 $\text{min} \rightarrow \text{max}$



min vayr

→ Median in a Stream



Max
siz $\rightarrow$ Min
Heap

atmost 1

Left
 $\downarrow$
Max
H

Right
 $\downarrow$
Min
Heap

$\rightarrow$ size $\Rightarrow$ size $\leftarrow \frac{Lp + up}{2} = \text{mid}$

$\rightarrow$ size $< 1 = \text{size}$

(1)
Max

Min

→ Median → 1

(1)
Max

(2)
Min

→ Median = $\frac{1+2}{2} = 1.5$

(1)
Max

(2)(3)
Min

→ Median → (2)

(1)(2)
Max

(3)(4)
Min

→ Median = $\frac{2+3}{2} = 2.5$

{ 1 } 3

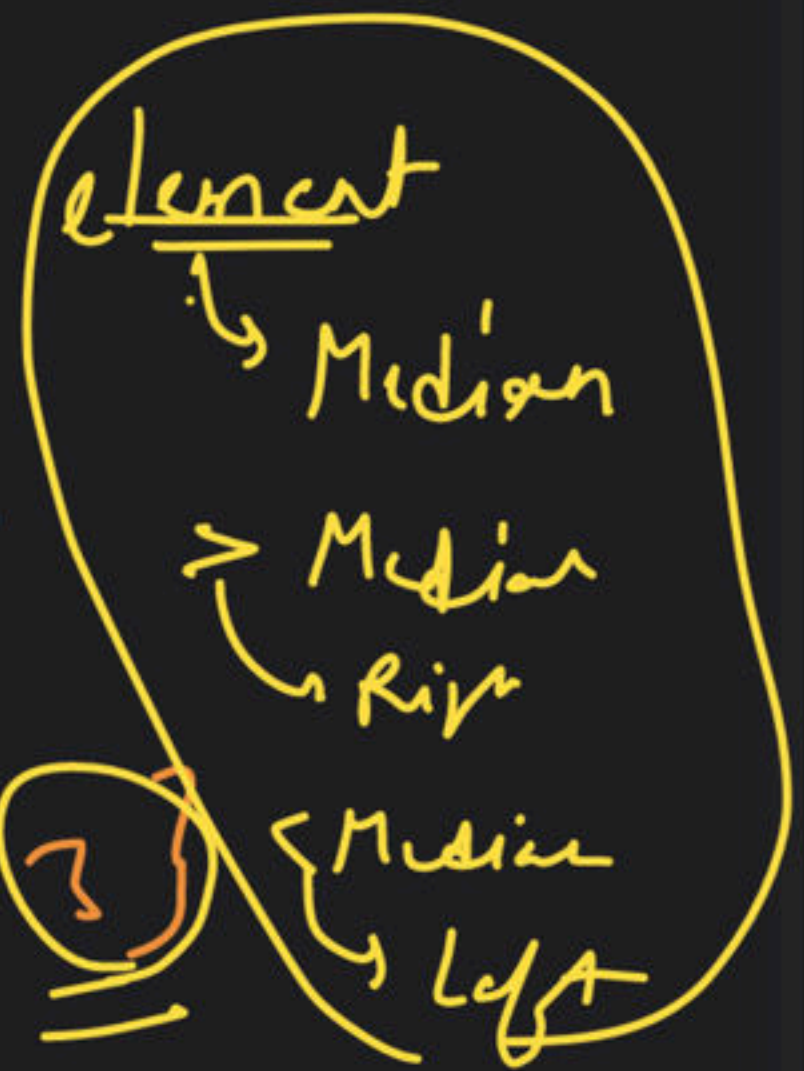
{ 1, 2 } 3

{ 1, 2, 3 }

{ 1, 2, 3, 4 }

{ 1, 2, 3, 4, 5 }

{ 1, 2, 3, 4, 5, 6 }



→ Max Size == Min Size

element → Median → > Median → Right

< Median → left

→ Max Heap == Min Heap Size + 1

element → Median → > Right → straight

< left → max top → Min
→ element - 1 Max

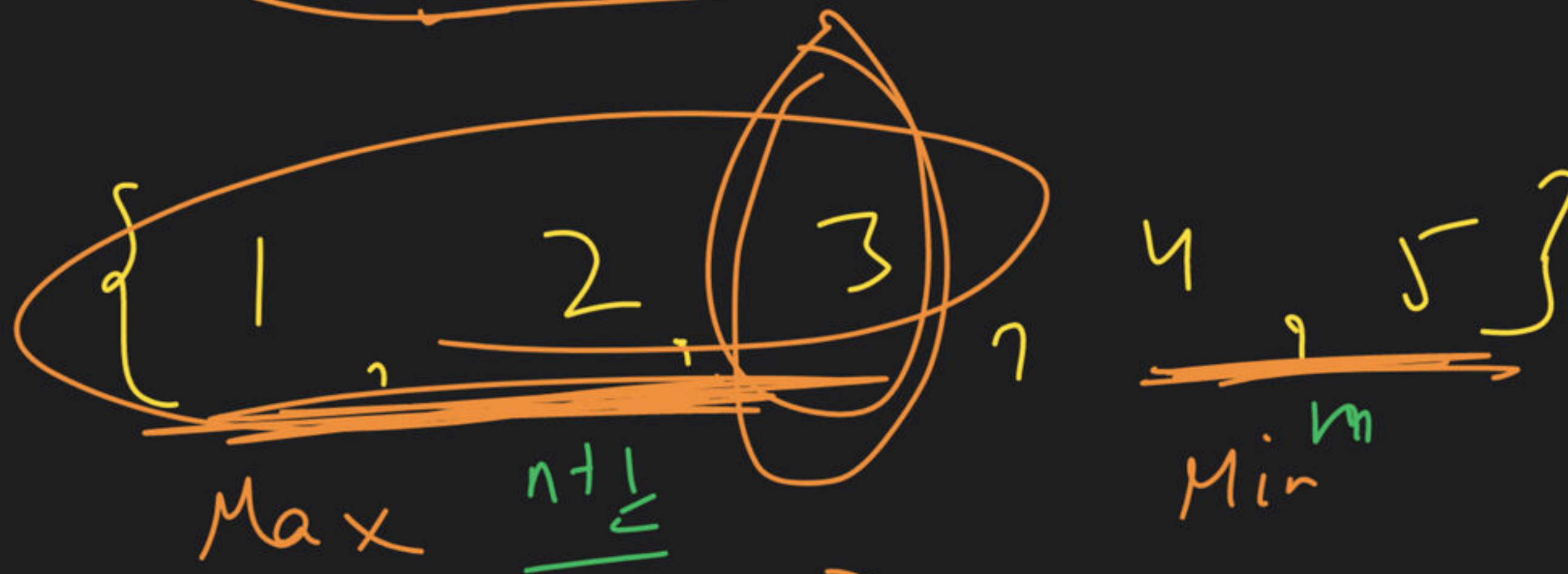
A

$$\rightarrow \text{MaxHeap} + 1 = \text{MinHeap}$$

$\text{element} > \text{Median} \rightarrow$
 $> \text{Right} \rightarrow \text{MinHeap} \rightarrow \text{max}$
 $\text{element} \rightarrow \text{insert}$

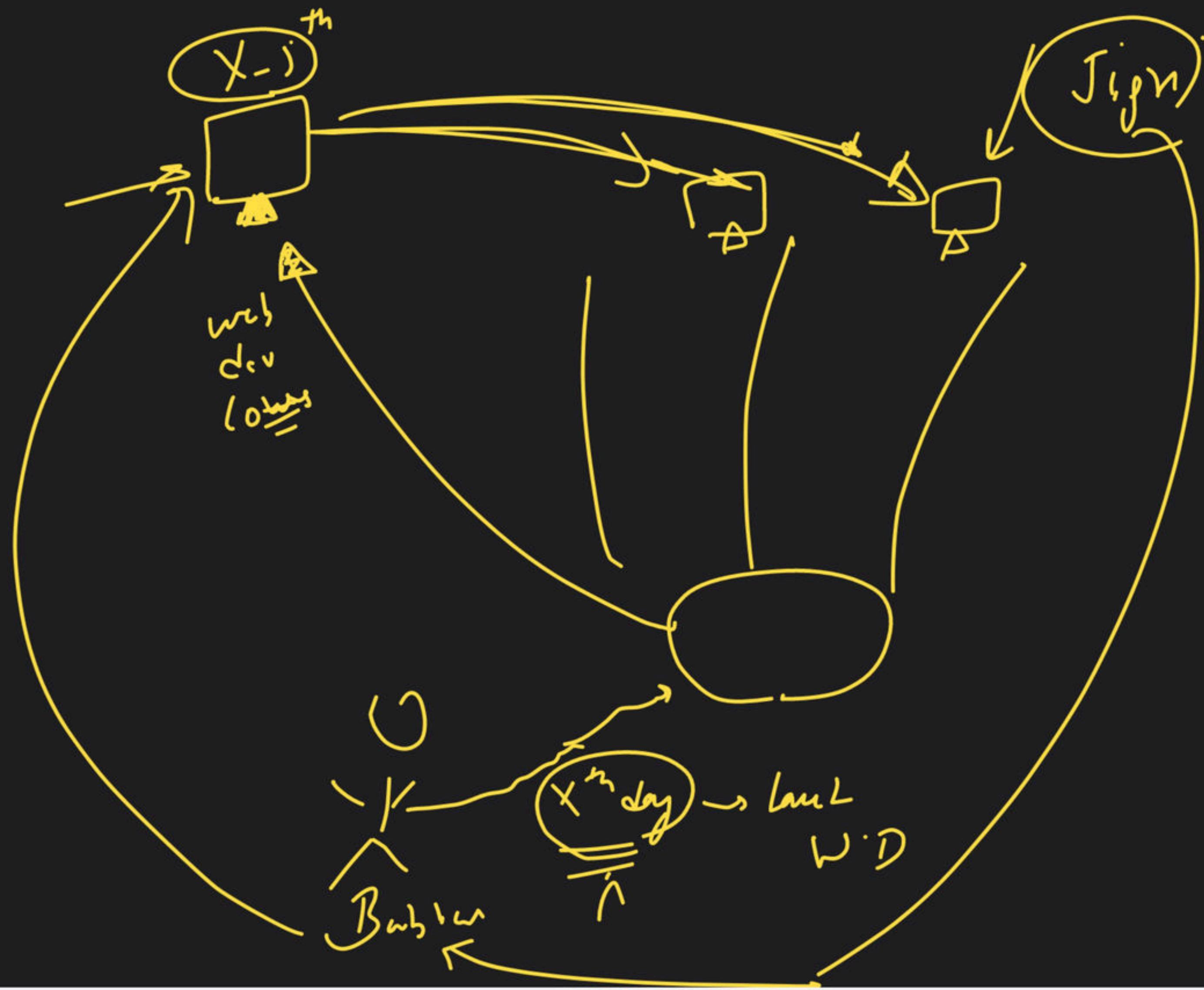
$< \text{Left} \rightarrow \text{straight}$

3 cars



{ 1, 2, 3, 4, 5, 6, 7 }

Max Min



Kal 8:30

Map trics

Q → Median in a Stream

